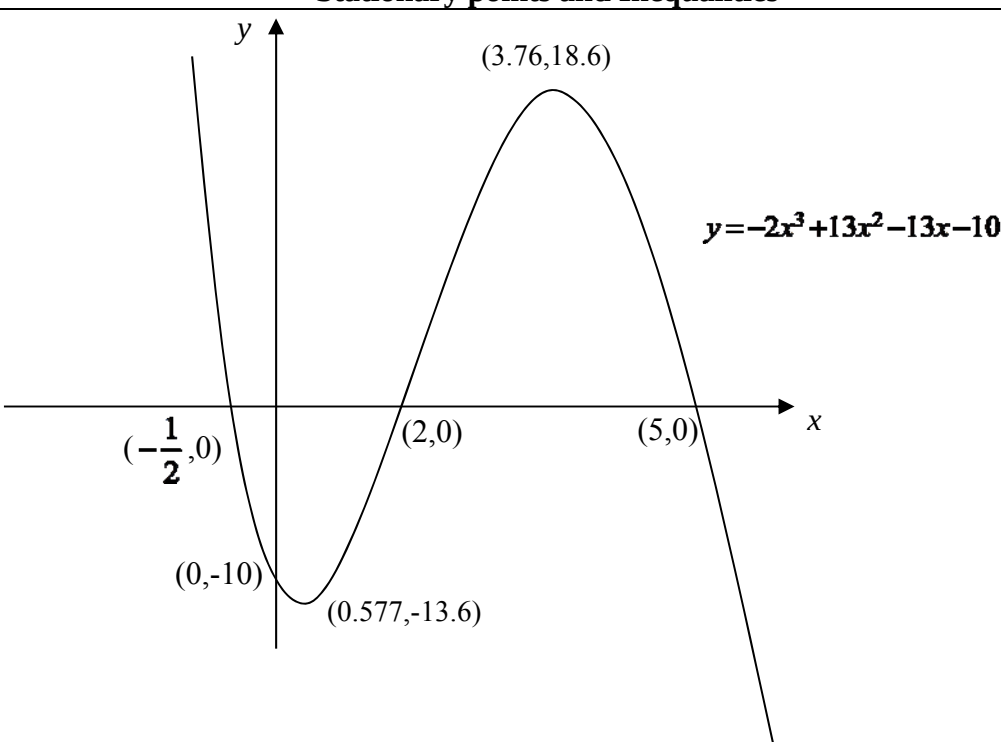


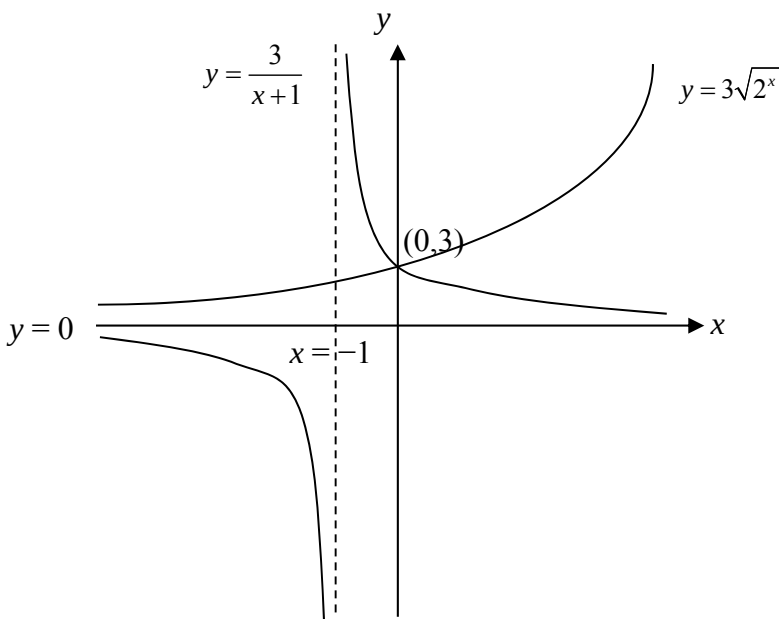
2016 H1 MATH (8864/01) JC 2 PRELIM EXAMINATION – MARKING SCHEME

Qn	Solution
1	Techniques of Integration
(a)	$\int \frac{e^{2x}(x-1)+2}{x-1} dx$ $= \int \left(e^{2x} + \frac{2}{x-1} \right) dx$ $= \frac{1}{2}e^{2x} + 2\ln x-1 + c$
(b)	$\int_0^a (3x-a)^2 dx$ $= \left[\frac{(3x-a)^3}{3(3)} \right]_0^a$ $= \frac{1}{9}(8a^3 + a^3)$ $= a^3$

Qn	Solution
2	Stationary points and Inequalities
(i)	
(ii)	For $-2x^3 + 13x^2 - 13x - 10 < 0$, $-\frac{1}{2} < x < 2 \quad \text{or} \quad x > 5$ For $-2e^{3x} + 13e^{2x} - 13e^x - 10 < 0$ Replace x with e^x in earlier part,

	$-\frac{1}{2} < e^x < 2 \quad \text{or} \quad e^x > 5$ Since $e^x > 0$, $0 < e^x < 2$ or $e^x > 5$ $\therefore x < \ln 2$ or $x > \ln 5$
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Qn	Solution								
3	Application of Differentiation								
(i)	Using Pythagoras Theorem, $RS = \sqrt{(2r)^2 - y^2}$ $\therefore P = 2QR + 2RS$ $= 2y + 2\sqrt{4r^2 - y^2}$ When P attains maximum value, $\frac{dP}{dy} = 2 + 2\left(\frac{1}{2}\right)(4r^2 - y^2)^{-\frac{1}{2}}(-2y) = 0$ $(4r^2 - y^2)^{-\frac{1}{2}} y = 1$ $y = (4r^2 - y^2)^{\frac{1}{2}}$ $y^2 = 4r^2 - y^2$ $2y^2 = 4r^2$ $y = \sqrt{2}r \text{ (reject } y = -\sqrt{2}r \text{ since } y > 0)$ $P = 2(\sqrt{2}r) + 2\sqrt{4r^2 - (\sqrt{2}r)^2} = 2\sqrt{2}r + 2\sqrt{2}r = 4\sqrt{2}r$ <table><tr><td>y</td><td>$(\sqrt{2}r)^-$</td><td>$\sqrt{2}r$</td><td>$(\sqrt{2}r)^+$</td></tr><tr><td>$\frac{dP}{dy}$</td><td></td><td></td><td></td></tr></table> Therefore P is a maximum when $y = \sqrt{2}r$.	y	$(\sqrt{2}r)^-$	$\sqrt{2}r$	$(\sqrt{2}r)^+$	$\frac{dP}{dy}$			
y	$(\sqrt{2}r)^-$	$\sqrt{2}r$	$(\sqrt{2}r)^+$						
$\frac{dP}{dy}$									

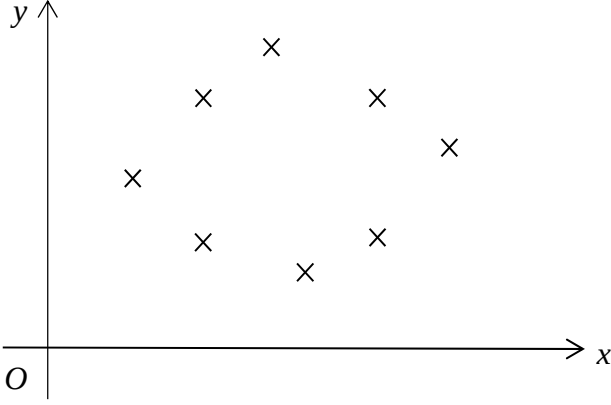
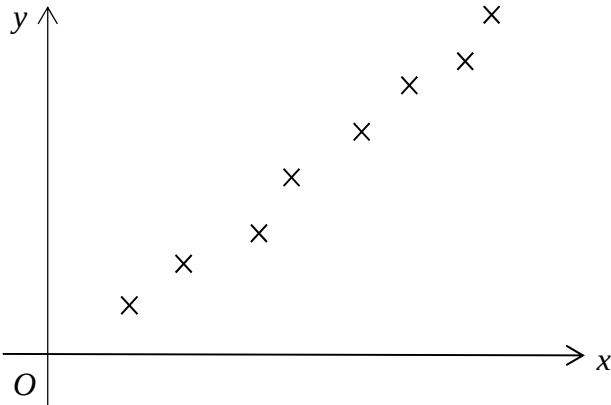
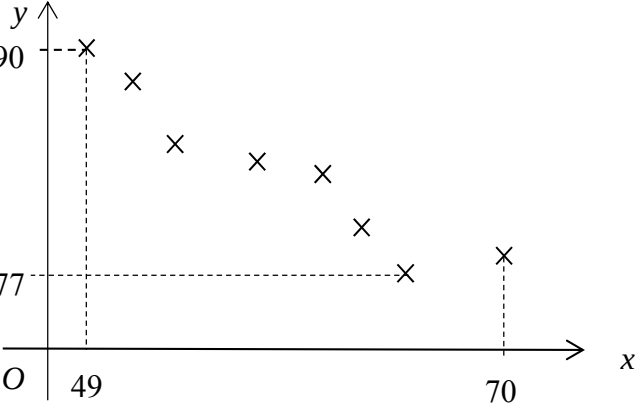
Qn	Solution
4	Curve Sketching + Application of Integration
(i)	
(ii)	$\sqrt{2^x} - \frac{1}{x+1} \geq 0$ $x < -1 \text{ or } x \geq 0$
(iii)	$\int_0^3 3\sqrt{2^x} - \frac{3}{x+1} dx = 11.668 \text{ (3 d.p.)}$

Qn	Solution
5	Applications of Differentiation (Tangent and Normal)
(i)	$y = \frac{1}{x}$ $\frac{dy}{dx} = -\frac{1}{x^2}$ At $x = a, y = \frac{1}{a}, \frac{dy}{dx} = -\frac{1}{a^2}$, Equation of tangent at A: $y - \frac{1}{a} = -\frac{1}{a^2}(x - a)$ $y = -\frac{1}{a^2}(x - a) + \frac{1}{a}$ $y = -\frac{1}{a^2}x + \frac{2}{a}$
(ii)	When $x = 0, y = \frac{2}{a}$ When $y = 0, \frac{1}{a^2}x = \frac{2}{a} \Rightarrow x = 2a$ Area of triangle $OPQ = \frac{1}{2}(2a)\left(\frac{2}{a}\right) = 2$ (independent of A)
(iii)	Equation of normal at A: $y - \frac{1}{a} = a^2(x - a)$ $y = a^2(x - a) + \frac{1}{a}$ $y = a^2x - a^3 + \frac{1}{a}$ Since the normal passes through the origin, $0 = a^2(0) - a^3 + \frac{1}{a}$ $\Rightarrow a^3 = \frac{1}{a}$ $\Rightarrow a = \pm 1$

Qn	Solution										
6	Sampling Methods										
(i)	1. Divide the employees in the company into mutually exclusive strata according to the department they work in. 2. <table><tr><td></td><td>Manufacturing Department</td><td>Assembly Department</td><td>Administrative Department</td><td>Total</td></tr><tr><td>Sample size</td><td>$100 \times \frac{450}{1000} = 45$</td><td>$100 \times \frac{300}{1000} = 30$</td><td>$100 \times \frac{250}{1000} = 25$</td><td>100</td></tr></table> 3. Using a Random Number Generator, randomly select 45 employees from the Manufacturing Department, 30 employees from the Assembly Department, and 25 employees from the Administrative Department. 4. The sample from each stratum is then combined to form a sample size of 100.		Manufacturing Department	Assembly Department	Administrative Department	Total	Sample size	$100 \times \frac{450}{1000} = 45$	$100 \times \frac{300}{1000} = 30$	$100 \times \frac{250}{1000} = 25$	100
	Manufacturing Department	Assembly Department	Administrative Department	Total							
Sample size	$100 \times \frac{450}{1000} = 45$	$100 \times \frac{300}{1000} = 30$	$100 \times \frac{250}{1000} = 25$	100							
(ii)	Using stratified sampling with the strata above will give a good representative sample as compared to random sampling because the working hours of employees from different departments have been considered and the sample size of each stratum is proportional to the size of the stratum. A better stratified sample of size 100 could have been achieved by dividing the employees into 6 strata according to their departments and level of seniority, as follows: Junior level employees in Manufacturing Department, Senior level employees in Manufacturing Department Junior level employees in Assembly Department, Senior level employees in Assembly Department Junior level employees in Delivery Department, Senior level employees in Delivery Department										

Qn	Solution
7	Probability
(i)	Required probability = $p + (1 - p)1.2p$ $= p + 1.2p - 1.2p^2$ $= 2.2p - 1.2p^2 \text{ (shown)}$ $2.2p - 1.2p^2 \geq 0.99$ Using GC, the smallest value of $p = 0.8$ (1 d.p.)
(ii)	$\frac{p}{2.2p - 1.2p^2} = 0.7$ $p = 1.54p - 0.84p^2$ $0.84p^2 - 0.54p = 0$ $p(0.84p - 0.54) = 0$ $p = 0 \text{ (rej) or } p = 0.643 \text{ (3 s.f.)}$
(iii)	Required probability = $0.7 \times (1 - 0.7)1.2(0.7) \times 2$ $= 0.3528$

Qn	Solution
8	Sampling
(i)	Let X be the price of 3 randomly chosen melons. Let Y be the price of a randomly chosen durian. $X = 5(M_1 + M_2 + M_3)$ $X \sim N(5 \times 3 \times 2.7, 5^2 \times 3 \times 0.3^2)$ $X \sim N(40.5, 6.75)$ $Y = 10D$ $Y \sim N(10 \times 1.5, 10^2 \times 0.1^2)$ $Y \sim N(15, 1)$ $P(X > 41 \text{ and } Y > 17) = P(X > 41)P(Y > 17)$ $= (0.42369)(0.02275)$ $= 0.0096389 \approx 0.00964 \text{ (3 s.f.)}$
(ii)	$X + Y \sim N(55.5, 7.75)$ $P(X + Y > 58) = 0.18459 \approx 0.185 \text{ (3 s.f.)}$
(iii)	Part (ii) includes cases that are not considered in part (i). For example, $X = 30$ and $Y = 28$ and $X = 48$ and $Y = 10$. In fact, cases in (i) form a proper subset of cases in (ii), which explains why the answer to part (ii) is greater than the answer to part (i).

Qn	Solution
9	Correlation and Regression
(a)(i)	
(a)(ii)	
(b)	Scatter diagram allows us to assess whether a linear relationship between the two variables exist first or The scatter diagram can be used to identify outliers, if they exist. or Even though $r = 0.993$ indicates a high positive linear correlation between the variables, x and t, the scatter diagram shows that as x increases, t increases at an increasing rate, hence a linear model may not be suitable.
(c)(i)	
b(ii)	$r = -0.958$

b(iii)	$y = -0.59530x + 117.90 \text{ (5 s.f)}$ $y = -0.595x + 118 \text{ (3 s.f)}$ When $x = 80$, $y = -0.59530(80) + 117.90$ $y = 70.276$ $y = 70.3 \text{ (3 s.f)}$ The estimate is unreliable as $x = 80$ is <u>outside the data range of x</u> and thus <u>linear relation between x and y may no longer hold.</u>
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Qn	Solution
10	Normal Distribution
(a)	If the mass of chickens has a normal distribution, there would be a possibility that a chicken has mass less than 0 kg, since $P(\text{mass of chicken} < 0) = 0.034454$. This means that 3.45% of the chickens would have mass less than 0 kg which is impossible.
(b) (i)	Let X be the daily milk production of a randomly chosen cow in litres $X \sim N(\mu, \sigma^2) \text{ and } \bar{X} = \frac{X_1 + X_2 + \dots + X_{16}}{16} \sim N\left(\mu, \frac{\sigma^2}{16}\right)$ $P(X > 35) = 0.25$ $1 - P(X \leq 35) = 0.25$ $P(X \leq 35) = 0.75$ $P\left(Z \leq \frac{35 - \mu}{\sigma}\right) = 0.75$ $\Rightarrow \frac{35 - \mu}{\sigma} = 0.67449$ $\Rightarrow \mu + 0.67449\sigma = 35 \quad \text{-----(1)}$ $P(\bar{X} \leq 25) = 0.089$ $P\left(Z \leq \frac{25 - \mu}{\frac{\sigma}{4}}\right) = 0.089$ $\Rightarrow \frac{4(25 - \mu)}{\sigma} = -1.3469$ $\Rightarrow 4\mu - 1.3469\sigma = 100 \quad \text{-----(2)}$ Using GC to solve (1) and (2), $\mu = 28.3 \quad (3 \text{ s.f.})$ $\sigma = 9.89 \quad (3 \text{ s.f.})$
(b) (ii)	Let Y be the number of cows, out of 19, that is grade A. $Y \sim B(19, 0.25)$ Required probability $= P(Y = 3)P(20\text{th cow is grade A})$ $= 0.15175 \times 0.25$ $= 0.0379 \quad (3 \text{ s.f.})$

Qn	Solution
11	Hypothesis Testing
(i)	Let $m = x + y$ $\sum m = 212.8 + 102.6 = 315.4 \quad \sum m^2 = 490.65 + 230.125 = 720.775$ Unbiased estimate of population mean using combined sample, $\bar{m} = \frac{315.4}{150} = \frac{1577}{750} = 2.1027 = 2.10 \text{ (3.s.f.)}$ Unbiased estimate of population variance using combined sample, $s_m^2 = \frac{1}{149} \left(720.775 - \frac{315.4^2}{150} \right) = 0.38654 = 0.387 \text{ (3 s.f.)}$
	Let M be the mass of a randomly chosen tin of milk powder. Let μ denote the population mean mass of a tin of milk powder. $H_0: \mu = 2$ $H_1: \mu > 2$ Since $n = 150$ is large, by Central Limit Theorem, $\therefore \bar{M} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$ approximately. Test statistic: $Z = \frac{\bar{M} - \mu}{S / \sqrt{n}}$ Level of Significance: 5% Reject H_0 if $p\text{-value} < 0.05$ Under H_0, using G.C., $p\text{-value} = 0.021565 = 0.0216 \text{ (3.s.f.)}$ Since $p\text{-value} = 0.0216 < 0.05$, we reject H_0 and conclude that at 5 % level of significance, there is sufficient evidence that the mean mass of a tin of milk powder is more than 2 kg. 0.0216 is the probability of obtaining a sample mean greater than or equal to 2.1027kg, assuming that the population mean mass of a tin of milk powder is 2kg.
	The null hypothesis is still rejected in favour of the alternative hypothesis, as $p\text{-value} = 2(0.021565) < 0.05$. Therefore there is sufficient evidence that the mean mass of a tin of milk powder is not 2 kg.
	Level of Significance: 5% Reject H_0 if $z\text{-value} > 1.6449$ $\frac{2.1027 - \mu_0}{\sqrt{0.38654} / \sqrt{150}} > 1.6449$ $\mu_0 < 2.1027 - (1.6449) \left(\frac{\sqrt{0.38654}}{\sqrt{150}} \right)$ $0 < \mu_0 < 2.02$

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Qn	Solution
12	Binomial Distribution
(a)(i)	Let X be the number of cookies, out of 12, that are chocolate cookies. $X \sim B(12, p)$ $E(X) = np = 5.4$ $\therefore p = \frac{5.4}{12} = 0.45$ $P(X = 3) = 0.092326$ $\approx 0.0923 \text{ (to 3 s.f.)}$
(ii)	$X \sim B(12, 0.45)$ Using GC, $P(X = 4) = 0.16996$ $P(X = 5) = 0.2225 \text{ (highest probability)}$ $P(X = 6) = 0.21238$ $\therefore$ most likely number is 5 cookies.
(b)(i)	Let Y be the number of cookies, out of n, that are chocolate cookies. $Y \sim B(n, 0.3)$ $P(Y \leq 8) \leq 0.9$ Using GC, When $n = 19$, $P(Y \leq 8) = 0.91608 > 0.9$ When $n = 20$, $P(Y \leq 8) = 0.88667 < 0.9$ $\therefore$ least n is 20.
(ii)	Total number of cookies = $12 \times 10 = 120$ Let W be the number of cookies, out of 120, that are chocolate cookies. $W \sim B(120, 0.3)$ Since n is large such that $np = 36 > 5$ and $n(1 - p) = 84 > 5$, $W \sim N(36, 25.2)$ approximately. $P(W > 30) = P(W > 30.5) \text{ after continuity correction}$ $= 0.863378$ $= 0.863 \text{ (to 3 s.f.)}$